FusionMVSA: Multi-View Fusion Strategy With Self-Attention for Enhancing Drug Recommendation

Yajie Meng, Zhuang Zhang, Xudong Shang, Xianfang Tang, Jincan Li, Zilong Zhang, Feifei Cui, Shuting Jin, Junlin Xu, and Peng Wang

***Abstract*—Leveraging the wealth of biomedical data available, we can derive insights into the relationships between biological entities from various angles. This underscores the complexity and significance of developing a dynamic approach for integrating data from multiple sources, a critical endeavor in drug recommendation. In this study, we introduce an innovative deep learning approach termed "Multi-View Fusion Strategy with Self-Attention" (FusionMVSA), designed to predict associations between drugs and diseases. To effectively amalgamate data from diverse sources and extract representative features, we have developed a feature extraction mechanism that capitalizes on similarities. This mechanism computes self-attention across multiple perspectives using shared group parameters, thereby highlighting common characteristics. Simultaneously, we utilize biomedical similarities among multi-source data as guiding factors for calculating similarity, enabling the capture of more nuanced features. Subsequently, we integrate these features through a feature fusion process, where known associations between drugs and diseases act as guiding terms. This strategy allows us to uncover the complementary aspects of different viewpoints. Ultimately, we predict potential drug-disease associations using a multi-layer perceptron neural network. Our methodology has undergone rigorous testing through various cross-validation experiments and case studies. We are confident that FusionMVSA will prove to be a valuable tool in drug recommendation, offering new avenues for exploration and discovery in the quest to combat diseases.**

***Index Terms*—Drug recommendation, drug–disease association, similarity-guided, multi-view, self-attention.**

Received 5 February 2025; revised 17 June 2025; accepted 30 June 2025. Date of publication 7 July 2025; date of current version 4 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62302156 and Grant 62402349, in part by the Natural Science Foundation of Hunan Province under Grant 2023JJ40180, in part by theNatural Science Foundation of Hubei Province under Grant 2024AFB127, in part by Wuhan Textile University Foundation under Grant 20230612 and Grant 2024309, and in part by-Hunan Province "Three Point" Science and Technology Innovation Plan Youth Science and Technology Talent Project under Grant2022RC1101. *(Corresponding authors: Junlin Xu; Peng Wang.)*

Yajie Meng, Zhuang Zhang, Xianfang Tang, and Jincan Li are with the School of Computer Science and Artificial Intelligence, Wuhan Textile University, Wuhan 430200, China (e-mail: myj@hnu.edu.cn; 2315363095@wtu.edu.cn; txf@hnu.edu.cn; 2215363099@wtu.edu.cn).

Xudong Shang is with the College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China (e-mail: xudongshang@hnu.edu.cn).

Zilong Zhang and Feifei Cui are with the School of Computer Science and Technology, Hainan University, Haikou 570228, China (e-mail: zhangzilong@hainanu.edu.cn; feifeicui@hainanu.edu.cn).

Shuting Jin and Junlin Xu are with the School of Computer Science and Technology, Wuhan University of Science and Technology, Wuhan 430065, China (e-mail: shutingjin@wust.edu.cn; xjl@hnu.edu.cn).

Peng Wang is with the School of Electronics and Information Engineering, Hunan First Normal University, Changsha 410002, China (e-mail: pwang@hnu.edu.cn).

Digital Object Identifier 10.1109/JBHI.2025.3586758

I. Introduction

DE NOVO drug development is notoriously costly, time-consuming, and fraught with risk. According to estimates by Scannell [1], the process of bringing a new drug to market typically spans 13 to 15 years, with average expenditures ranging from two to three billion USD. Despite these substantial investments, most drug development efforts fail due to various issues such as ineffectiveness or toxicity [2]. Therefore, identifying low-risk and efficient methods for drug development is both meaningful and critical.

Drug recommendation is an effective approach to enhance therapeutic outcomes and shorten the drug development cycle. Among the various process involved in drug recommendation, identifying novel drug-disease associations is a critical step. With the accumulation and availability of life sciences databases, such as OMIM [3], DrugBank [4], PubChem [5], DrugCentral [6], and HDVD [7], numerous computational methods have been proposed over the past decade to predict drug-disease associations. These predictions can serve as valuable guiding information for subsequent experimental validation.

The first widely adopted computational framework for drug recommendation was based on a network analysis. DrugNet [8] is a network of interconnected drugs, proteins, and diseases, where associations are predicted through a network-based prioritization method. MBiRW [9] integrates a drug similarity network and a disease similarity network into a heterogeneous network with known drug-disease interactions, employing a bi-random walk algorithm to predict potential associations.

Machine learning-based methods have been widely applied to various bioinformatics tasks [10], [11]. In drug-disease association prediction, these methods typically leverage diverse biological information about drugs and diseases as prior knowledge,

2168-2194 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.

modeling the association prediction as a binary classification problem [12], [13], [14], [15], [16], [17], [18], [19], [20]. For example, PREDICT [21] gathers various data, such as drug molecular activity, drug side effects, and disease phenotypes, and applies a logistic regression classifier to identify new drug-disease associations. LRSSL [22] integrates drug chemical information, drug target domain information, and target annotation information to predict new associations using Laplacian regularized sparse subspace learning. However, these feature-based classification methods heavily rely on the extraction of features and the selection of negative samples.

In addition to traditional machine learning techniques, matrix completion and matrix factorization have been successfully adopted to predict drug-disease associations. For instance, SCMFDD [23] projects drug-disease association relationships into two latent low-rank spaces under the constraints of biological similarity and employs a non-negative matrix factorization method on these spaces to predict new drug-disease associations. HGIMC [24] utilizes multiple similarities of drugs and diseases, predicting potential indications via a heterogeneous graph inference method based on matrix completion. It is important to note that deploying these machine learning methods on large-scale data is challenging due to the high complexity of matrix operations involved.

As an important subfield of machine learning, deep learning-based methods have proven useful for drug-disease association prediction. For example, BiFusion [25] is a bipartite graph convolutional network that uses proteins as a bridge for message passing between drug space and disease space. It feeds the extracted features of the drug and disease into a multi-layer perceptron decoder to model drug-disease associations. DRHGCN [26] extracts features via a heterogeneous information fusion graph convolutional network, which fuses inter-domain and intra-domain features in parallel, and then applies an inner-product decoder to predict the scores of drug-disease associations. These approaches typically utilize only a single similarity measure between pairwise diseases and pairwise drugs. However, the similarities between pairwise drugs (or pairwise diseases) are often noisy, multi-modal, and can be measured from different aspects. By fusing multiple similarity measures from various drug and disease data sources, different representations of the data can be captured, thus further improving the model's predictive ability.

Due to the shortcomings of the aforementioned methods, we propose a novel deep learning-based model,termed the Multi-View Fusion Strategy with Self-Attention(FusionMVSA), aimed at enhancing the prediction of drug-disease associations. Unlike traditional methods, FusionMVSA incorporates a novel self-attention mechanism as a feature aggregation method instead of the commonly used graph convolutional network (GCN) [27], [28],allowing for a more comprehensive fusion of data across various modalities. Specifically, we introduce two innovative approaches to compute self-attention for the biological bipartite network and design a feature extraction component and a feature fusion component based on these computational methods.

TABLE I
SUMMARY OF THE TWO DATASETS

Datasets	Drugs	Diseases	Associations	LD*
Fdataset	593	313	1933	1.04e-2
Cdataset	663	409	2532	0.87e-2

Note: LD* is the link density, which is defined as the ratio of known associations to all possible associations.

II. MATERIALS AND METHOD

A. Problem Definition

Predicting new drug-disease associations can be viewed as a binary classification task within a biological bipartite network. This network consists of two distinct sets of biological entities, one for n drugs, denoted as $R = r_1, r_2, \ldots, r_n$, and another for m diseases, denoted as $D = d_1, d_2, \ldots, d_m$. The associations in this network are represented by a binary adjacency matrix $\mathbf{Y} \in \mathbb{R}^{n \times m}$. Specifically, the element $\mathbf{Y}_{i,j}$ is set to 1 if drug r_i is associated with disease d_j and this association is observed, and 0 otherwise. An entry of 0 indicates that the corresponding drug and disease are either not associated or the association is unobserved. In this paper, we propose FusionMVSA, a method designed to uncover novel associations by leveraging multiple biological similarities between drugs and diseases, alongside known drug-disease associations.

B. Datasets

In our study, we conducted experiments on two benchmark datasets: Fdataset [21], and Cdataset [9].The Fdataset, a widely accepted gold standard dataset, comprises 593 drugs, 313 diseases, and 1,933 known associations. The Cdataset is formed by combining the DNdataset [8] and Fdataset, resulting in a dataset with 663 drugs, 409 diseases, and 2,352 known associations. The detailed statistics of these two datasets are provided in Table I.

C. Drug-Drug Similarity

One guiding intuition in drug recommendation is that similar drugs tend to associate with similar diseases. To quantify this similarity, a drug's characteristics can be encoded using a binary vector, where each bit, set to 1 or 0, represents a unique attribute. Then, similarity between two drugs, represents by feature $\mathcal{P}$ and $\mathcal{Q}$, is typically calculated by the Jaccard score [29], [30] as follows:

$$Jaccard(\mathcal{P}, \mathcal{Q}) = \frac{|\mathcal{P} \cap \mathcal{Q}|}{|\mathcal{P} \cup \mathcal{Q}|} \tag{1}$$

In the Jaccard score calculation, the symbols $\cap$ and $\cup$ denote the bit-wise AND and bit-wise OR operations, respectively. Additionally, the symbol $\|$ signifies the total count of bits with a value of 1in the specified vector.

In recent biological research endeavors, methodologies analogous to those of Wang et al. [20], which incorporate multivariate data as model inputs, have seen a notable surge in adoption.

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SUMMARY OF SIMILARITY OF DRUGS AND DISEASES

Spaces	Notations	Descriptions
Drug	S^R_{GS}	Chemical structure similarity
	S^R_{DD}	Drug-drug interactions similarity
	S^R_{DT}	Drug-target information similarity
	S^R_{SE}	Side effect similarity
Disease	S^D_{DP}	Disease phenotype similarity
	S^D_{DO}	Disease ontology similarity

Consequently, in this study, we consider each type of feature as a distinct view and calculate drug similarity from four different perspectives. This approach aims to enhance the efficacy of detecting novel drug-disease associations. The four views are as follows: (1) Chemical structure similarity: The Simplified Molecular Input Line Entry Specification (SMILES) [31] is a linear character string that describes the chemical molecular compound. By extracting the canonical SMILES of all drugs from the DrugBank database, we employ a Chemical Development Kit (CDK) [32] to generate binary fingerprints for the drugs. We then calculate the chemical structure similarity based on these fingerprints. (2) Drug-drug interactions indicate the potential impact when combining two drugs for treatment. This similarity measure is determined by the Jaccard scores of their interaction profiles, which can be sourced from DrugBank. (3) Drug–target information similarity: The target of a drug refers to the binding site between the drug and biological macromolecules. We obtain the drug-target information from DrugBank and calculate the drug-target information similarity using Jaccard scores. (4) Side effect similarity: Information regarding drug side effects is crucial for our analysis. This data is sourced from the Side database [33], which encompasses 1,430 drugs along with their respective side effects. We encode these side effects as a binary vector and determine the Jaccard similarity based on this information.

D. Disease-Disease Similarity

We gather two perspectives on disease similarity as outlined below: (1) This data is retrieved from the MimMiner [34] database, which represents the disease phenotype vector by employing text mining on the MeSH terms [35] and calculates the cosine similarity of their angles.(2) Disease ontology similarity: This information is acquired from HGIMC [24],which utilizes a Gene Ontology-based algorithm [36] on the structure of disease ontology terms.

The similarity descriptors for the drug space and disease space can be found in Table II.

III. MODEL ARCHITECTURE

This section elaborates on FusionMVSA for predicting drug-disease associations. The comprehensive workflow of FusionMVSA is illustrated Fig. 1.

A. GCN and Self-Attention

A graph convolutional network (GCN) [27] is an efficient neural network architecture for processing graph structure data. Let $G \in \mathbb{R}^{h\times h}$ represent the adjacency matrix of a graph, and $H \in \mathbb{R}^{h\times d}$ represent the input embeddings matrix. The GCN aggregates features using the following formula:

$$\mathrm{GCN}(H, G) = \sigma\left(D^{-\frac{1}{2}} G D^{-\frac{1}{2}} H W\right) \tag{2}$$

where $D = \mathrm{diag}(\sum_j G_{i,j})$, and $\sigma(\cdot)$ is a ReLU [37] activation function. As a powerful feature aggregator in deep learning, GCNs have been widely used in biological bipartite networks. However, they can only aggregate the features of first neighbors in the first convolutional layer. To allow nodes to aggregate features from higher-order neighbors, many convolutional layers must be stacked, which can lead to oversmoothing [36]. In contrast, self-attention is a weighted global information aggregation approach. Initially used in machine translation [38], it has been widely adopted in the sequence data field, such as for learning the embeddings of words in sentences.

To calculate the self-attention of H, three matrices, $W_Q, W_K \in \mathbb{R}^{d\times d_k}$, and $W_V \in \mathbb{R}^{d\times d_v}$, are first employed to project H into different representation spaces as follows:

$$Q = HW_Q, \quad K = HW_K, \quad V = HW_V \tag{3}$$

Then, Q and K are used to compute the scaled dot-product similarity matrix, and obtain the aggregate weight A as below:

$$A = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) \tag{4}$$

where d_k is a scale factor, and is usually set as the feature dimension of K. softmax$(\cdot)$ is a normalization function, and $A_{i,j}$ is the weight when element i aggregates features of element j.

Finally, the self-attention of H is computed with the following formula:

$$\text{self-attention}(H) = AV \tag{5}$$

Compared to GCNs, the self-attention mechanism allows nodes to aggregate features from all other nodes in a single forward propagation. Furthermore, the aggregate weights are dynamically calculated using scaled dot-product similarity. Therefore, we introduced self-attention into the biological bipartite network and considered it the primary structure for feature aggregation.

The drug space similarities and disease space similarities were padded with zeros, and the initial embeddings of different views were computed as follows:

$$E^R_i = \begin{bmatrix} S^R_i & \text{zeros} \end{bmatrix} \in \mathbb{R}^{n\times(n+m)} \tag{6}$$

$$E^D_j = \begin{bmatrix} \text{zeros} & S^D_j \end{bmatrix} \in \mathbb{R}^{m\times(n+m)} \tag{7}$$

where $i = \{1, 2, \ldots, N_r\}$ and $j = \{1, 2, \ldots, N_d\}$. N_r and N_d represent the data source number of drug space and disease space; in this study, they are set to 4 and 2, respectively. Therefore, E^R_i and E^D_j represent the initial embeddings matrix of drugs and diseases.

B. Feature Extraction Component

To extract and integrate features from multiple views, we proposed a novel approach to compute self-attention and designed a

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.

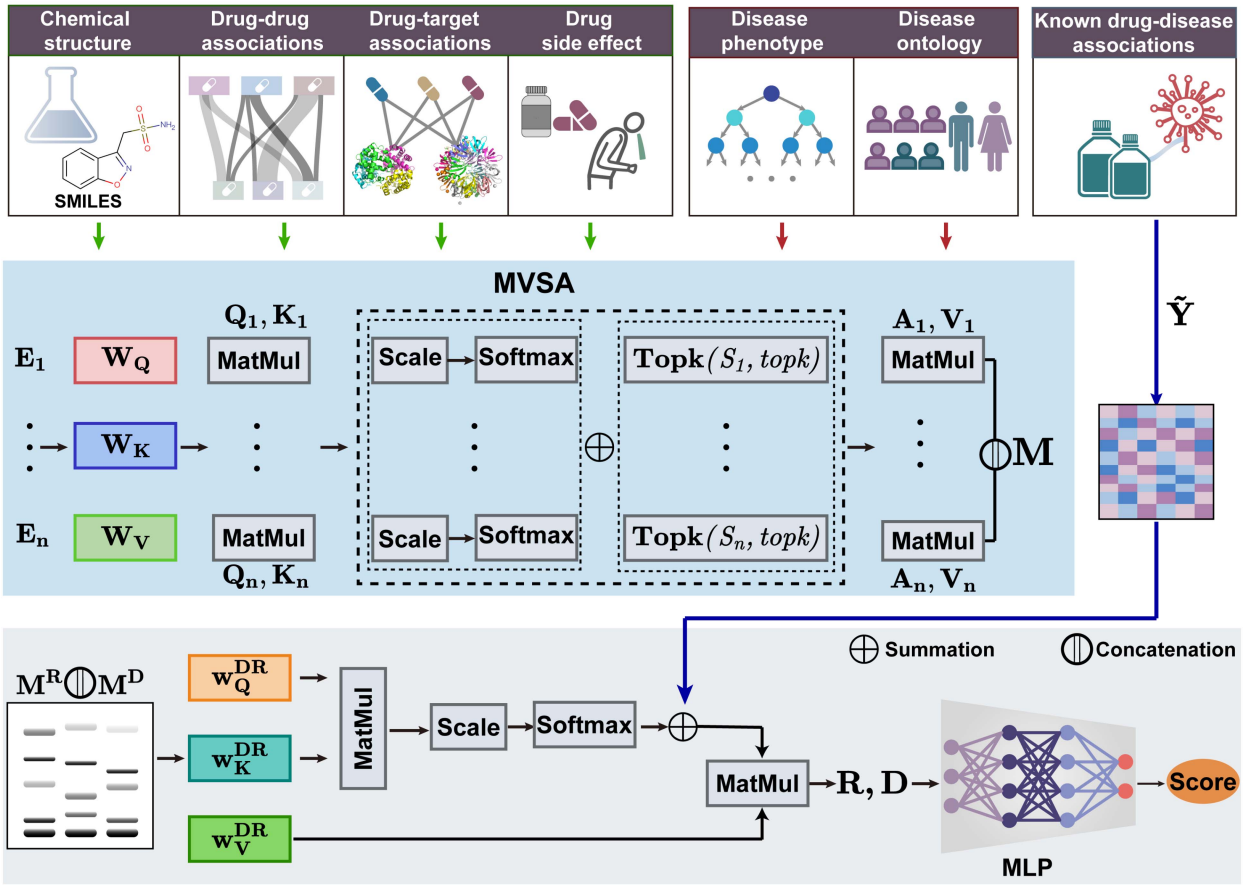


Fig. 1. Overall architecture of FusionMVSA, which consists of three main components: (i) a multi-view self-attention mechanism extracts representative features from multi-source drug and disease data; (ii) to prevent data leakage, only known drug-disease associations from the training set are incorporated into the self-attention computation to fuse the drug and disease features; and (iii) a multi-layer perceptron (MLP) decoder processes the fused features to predict the final associations.

corresponding feature extraction component. FusionMVSA employs two feature extraction components in parallel to separately extract features from the drug space and the disease space. The workflow of the feature extraction component in the drug space is illustrated in Fig. 1.

Specifically, we first apply layer normalization [39] to the input embeddings to facilitate more effective optimization [40] Then, the Q, K, and V matrices for different views in the drug space and disease space are computed using the following formulas:

$$\begin{aligned} Q_i^R &= \mathrm{LN}(E_i^R)W_Q^R, \\ K_i^R &= \mathrm{LN}(E_i^R)W_K^R, \\ V_i^R &= \mathrm{LN}(E_i^R)W_V^R \end{aligned} \tag{8}$$

$$\begin{aligned} Q_j^D &= \mathrm{LN}(E_j^D)W_Q^D, \\ K_j^D &= \mathrm{LN}(E_j^D)W_K^D, \\ V_j^D &= \mathrm{LN}(E_j^D)W_V^D \end{aligned} \tag{9}$$

where $\mathrm{LN}(\cdot)$ represents the layer normalization function. $W_Q^R, W_K^R, W_V^R \in \mathbb{R}^{(n+m)\times(n+m)//N_r}$ and $W_Q^D, W_K^D, W_V^D \in \mathbb{R}^{(n+m)\times(n+m)//N_d}$ are two group parameters to project the normalized embeddings, and $i = \{1, 2, \ldots, N_r\}$, $j = \{1, 2, \ldots, N_d\}$. Taking Q_i^R as an example, the superscript R represents the embeddings was calculated in drug space, and the subscript i represents the i-th view.This notation for superscripts and subscripts is used consistently throughout this article.

Unlike multi-head self-attention (MHSA) [38], [41], which uses multiple group parameters, our approach employs a single group parameter to emphasize the common properties of different views within the same space. We refer to this as the multi-view self-attention (MVSA) mechanism.

Additionally, inspired by Graphormer [42], we introduce the attribute information of node pairs into the computation of self-attention. In this study, we consider four types of drug similarities and two types of disease similarities as important guiding terms for the aggregate weights. The final aggregate weights for different views are denoted as A_i^R or A_j^D, and are computed as below:

$$A_i^R = \mathrm{softmax}\left(\frac{Q_i^R \left(K_i^R\right)^\top}{\sqrt{d_{K_i}^R}}\right) + \mathrm{Topk}\left(S_i^R, a\right) \tag{10}$$

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.

$$A_j^D = \text{softmax}\left(\frac{Q_j^D \left(K_j^D\right)^\top}{\sqrt{d_{K_j}^D}}\right) + \text{Topk}\left(S_j^D, a\right) \quad (11)$$

where $d_{K_i}^R$ and $d_{K_j}^D$ are the feature dimensions of K_i^R and K_j^D, respectively, and $\text{Topk}(\cdot)$ is a strategy to control the influence of similarity-guided terms. Specifically, for similarity matrix S, if the element i is the first a nearest neighbor of the element j, the similarity value $S_{i,j}$ will be retained; otherwise, $S_{i,j}$ will be set as 0.

The features in different views of drug space and disease space are horizontally concatenated and then projected with W_{out}^R and W_{out}^D to restore them to their original dimensions $\mathbb{R}^{m\times m}$ or $\mathbb{R}^{n\times n}$. The multi-view self-attention (MVSA) of drugs and diseases are computed as follows:

$$\text{MVSA}(E^R) = \text{Concat}\left(A_1^R V_1^R, \ldots, A_{N_r}^R V_{N_r}^R\right) W_{\text{out}}^R \quad (12)$$

$$\text{MVSA}(E^D) = \text{Concat}\left(A_1^D V_1^D, \ldots, A_{N_d}^D V_{N_d}^D\right) W_{\text{out}}^D \quad (13)$$

Finally, we conduct layer normalization of the extracted features, and connect them with the mean of different views input embeddings. The final outputs of the feature extraction component are computed with the following formulas:

$$\text{Extraction}\left(E^R\right) = \frac{1}{N_r}\sum_{i=1}^{N_r} E_i^R + \text{LN}\left(\text{MVSA}(E^R)\right) \quad (14)$$

$$\text{Extraction}\left(E^D\right) = \frac{1}{N_d}\sum_{j=1}^{N_d} E_j^D + \text{LN}\left(\text{MVSA}(E^D)\right) \quad (15)$$

C. Feature Fusion Component

A feature fusion component based on self-attention is designed to fully integrate the drug features and disease features. The workflow of the feature fusion component is illustrated in Fig. 1.

The input embeddings of the feature fusion component are obtained by concatenating the integrated features of the two spaces as below: The input embeddings of the feature fusion component are obtained by concatenating the integrated features of the two spaces as below:

$$M = \begin{bmatrix}\text{Extraction}(E^R)\\ \text{Extraction}(E^D)\end{bmatrix} \quad (16)$$

As in the feature extraction component, a layer normalization operation is adopted here. Q, K and V are computed with W_Q^{RD}, W_K^{RD}, $W_V^{RD} \in \mathbb{R}^{(n+m)\times(n+m)}$ as follows:

$$\begin{aligned} Q^{RD} &= \text{LN}(M)W_Q^{RD},\\ K^{RD} &= \text{LN}(M)W_K^{RD},\\ V^{RD} &= \text{LN}(M)W_V^{RD} \end{aligned} \quad (17)$$

The known associations information Y is crucial to the prediction. To incorporate it into the computation of self-attention, Y is transformed to $\widetilde{Y} \in \mathbb{R}^{m\times(n+m)}$. Specifically, the first n rows of $\widetilde{Y}$ represent drugs, and the remaining m rows represent diseases. The element of $\widetilde{Y}_{i,j}$ is set to 1 if the corresponding drug and disease are associated with each other; it is also set to 1 if i = j. This transformation proceeds as follows:

$$\widetilde{Y}_{i,j} = \begin{cases} 1 & \text{if } Y_{i,(j-n)} = 1 \text{ or } Y_{j,(j-m)} = 1 \text{ or } i = j,\\ 0 & \text{otherwise.}\end{cases} \quad (18)$$

Clearly, $\widetilde{Y}$is a symmetric matrix. We make $\widetilde{Y}$ be additional guided information for self-attention by combining it with the aggregate weights computed by the scaled dot-product similarity. Therefore, the features of different spaces are fused with the below formulas:

$$A^{RD} = \text{softmax}\left(\frac{Q^{RD}\left(K^{RD}\right)^\top}{\sqrt{d_{K^{RD}}}}\right) + \widetilde{Y} \quad (19)$$

$$\text{SA}(M) = A^{RD}V^{RD}W_{\text{out}}^{RD} \quad (20)$$

where $d_{K^{RD}}$ is the feature dimension of K^{RD}, and $W_{\text{out}}^{RD} \in \mathbb{R}^{(n+m)\times(n+m)}$.

The final outputs of the feature fusion component are computed as:

$$\begin{bmatrix}R\\ D\end{bmatrix} = M + \text{LN}(\text{SA}(M)) \quad (21)$$

where R and D represent the final features of all drugs and diseases, respectively.

To prevent model overfitting and reduce noise from laboratory measurement data, regular dropout [43] is applied on A_i^R, A_j^D, and A^{RD} when computing self-attention.

D. Decoder

A multi-layer perceptron neural network decoder is employed to reconstruct the associations in the drug-disease bipartite network. For drug r_i and disease d_j, we feed the corresponding feature pair $(R_i, D_j) \in \mathbb{R}^{1\times(n+m)}$ into a three-layer linear neural network as follows:

$$\text{MLP1} = \text{Relu}(\text{Linear}(R_i, D_j)), \quad \text{MLP1} \in \mathbb{R}^{1\times(n+m)} \quad (22)$$

$$\text{MLP2} = \text{Relu}(\text{Linear}(\text{MLP1})), \quad \text{MLP2} \in \mathbb{R}^{1\times\frac{(n+m)}{2}} \quad (23)$$

$$P_{i,j} = \text{Sigmoid}(\text{Dropout}(\text{MLP2})) \quad (24)$$

where $P_{i,j}$ represents the possibility of association between drug r_i and disease d_j.

E. Optimization

Known drug–disease associations are essential for predicting novel associations. We denote the pair of drug r_i and disease d_j as (i,j), the known association pairs as y^+, and the set of other pairs as y^-. Here, a weighted binary cross-entropy function is adopted as our loss function:

$$\text{loss} = -\frac{1}{n\times m}\left(\lambda \times \sum_{(i,j)\in\mathcal{Y}^+} \log P_{i,j} + \sum_{(i,j)\in\mathcal{Y}^-} \log(1 - P_{i,j})\right) \quad (25)$$

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1: FusionMVSA.

Input: multiple drug similarities $\{S_1^R, S_2^R, \ldots, S_{N_r}^R\} \in \mathbb{R}^{N_r \times n \times n}$;multiple disease similarities $\{S_1^D, S_2^D, \ldots, S_{N_d}^D\} \in \mathbb{R}^{N_d \times m \times m}$;known associations $\mathbf{Y} \in \mathbb{R}^{n \times m}$; parameter $topk$; training epochs N

Output: prediction matrix $\mathbf{Y}^*$.

Process the input data: initialize multi-view embedding of drugs and diseases using (6) and (7), transform $\mathbf{Y}$ with (18), and conduct forward propagation.

For $I = 1$ to N:

1) Extract drug space feature from multi-view embedding with (14).
2) Extract disease space feature from multi-view embedding with (15).
3) Fuse the features of drug space and disease space with (21).
4) Decode drug-disease feature pair, and compute associate possibility with (24).
5) Calculate loss with (25), and update parameters using the Adam optimizer.

Output: $\mathbf{Y}^*$.

where $\lambda = \frac{|\mathcal{Y}^-|}{|\mathcal{Y}^+|}$ is the balance factor that imposes the influence of the observed drug–diseases associations, and $|\mathcal{Y}^-|$ and $|\mathcal{Y}^+|$ are the number of pairs in $\mathcal{Y}^-$ and $\mathcal{Y}^+$, respectively.

The loss function is minimized using an Adam [44] optimizer, which updates the parameters iteratively. The detailed steps of FusionMVSA are described in Algorithm 1.

F. Time Complexity Analysis

The computational overhead of the proposed model primarily arises from three components: feature extraction, feature fusion, and decoding. For feature extraction, the time complexity for extracting features from the drug and disease semantic views is $O(N_r \times d \times |V_{DR}|^2)$ and $O(N_d \times d \times |V_{DI}|^2)$, respectively, where N_r and N_d represent the number of drug and disease views, d is the embedding dimension, and $|V_{DR}|$ and $|V_{DI}|$ denote the number of drugs and diseases. The feature fusion component calculates the drug-disease interactions using self-attention, with time complexity $O(d \times D^3)$, where $D = |V_{DR}| + |V_{DI}|$. The decoding phase, used for predicting drug-disease pairs, has a time complexity of $O(B \times d \times D^2)$, where B is the number of drug-disease pairs to be predicted.

Thus, the overall time complexity of the model is:

$$O(d \times (N_d \times |V_{DR}|^2 + N_r \times |V_{DI}|^2 + D^3 + B \times D^2))$$

The primary computational bottleneck occurs in the interaction view feature fusion stage, where the complexity is $O(D^3)$. This becomes particularly expensive when the number of drugs and diseases is large. In prediction scenarios where $B < D$, interaction modeling dominates the computational cost; conversely, when $B > D$, the decoding phase becomes the primary computational cost.

IV. Result

Ten-fold cross-validation is performed on the F dataset and C dataset to compare the performance of FusionMVSA with other existing methods. Subsequently, an ablation analysis is conducted to verify the effectiveness of the guided terms in FusionMVSA. Finally, case studies on 10 neoplasms are carried out to demonstrate the usefulness of FusionMVSA in drug recommendation.

A. Experiment Setting

In the experiment, we employ the K-fold cross-validation approach, a widely used framework that comprehensively evaluates prediction performance by minimizing the influence of different dataset splits. We select five evaluation metrics: the area under the receiver operating characteristic curve (AUC), area under the precision-recall curve (AUPR), precision, recall, and F1-score. The F1-score is a comprehensive metric that combines precision and recall. AUPR and the F1-score are particularly important in this study because the known and unknown associations in the two benchmark datasets are highly imbalanced.

To calculate the evaluation metrics, we set a threshold θ. When the drug–disease association prediction score is higher than θ, the prediction is identified as a positive sample; otherwise, it is identified as a negative sample. The true positive rate (TPR), false positive rate (FPR), precision, recall, and F1-score are calculated using the following formulas:

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{26}$$

$$\text{FPR} = \frac{\text{FP}}{\text{TN} + \text{FP}} \tag{27}$$

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{28}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{29}$$

$$\text{F1} = 2 \times \frac{\text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \tag{30}$$

All experiments were conducted on a workstation equipped with a single NVIDIA RTX 4090 GPU (24 GB). We implemented our models using PyTorch v1.13.1 and Python v3.9.21, with CUDA v11.8 enabled for GPU acceleration. In our study, we set the dropout rate to 0.1, the learning rate to 1e-3, and the number of training epochs to 800 for all experiments.

B. Parameter Evaluation

The parameter α is a shared factor in the two feature extraction components, controlling the influence of guided terms. We considered the range of α as 2,3,...,11. Experiments are carried out on the gold-standard Fdataset, and evaluated under 10-fold cross-validation.

As shown in Table III, in the value of topk increased from 2 to 11, the values of AUC, AUPR, and F1-score initially increased and then decreased. This trend indicates that similarity-guided terms with appropriately large values are beneficial for

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.

TABLE III
PERFORMANCE OF DIFFERENT TOPK BASED ON 10-FOLD CROSS-VALIDATION ON FDATASET

topk	AUC	AUPR	Precision	Recall	F1-score
2	0.9427	0.6056	0.7011	0.5396	0.6077
3	0.9451	0.6068	**0.7156**	0.5354	0.6115
4	0.9479	0.6076	0.6882	0.5613	0.6163
5	0.9488	0.6078	0.7003	0.5469	0.6124
6	0.9505	0.6159	0.6976	0.5571	0.6182
7	**0.9514**	**0.6178**	0.6661	**0.5799**	**0.6192**
8	0.9504	0.6115	0.7082	0.5421	0.6107
9	0.9495	0.6119	0.6698	0.5665	0.6091
10	0.9499	0.6087	0.6676	0.5623	0.6089
11	0.9488	0.6036	0.6971	0.5334	0.6018

Note: The best results in each row are in bold faces and the second-best results are underlined.

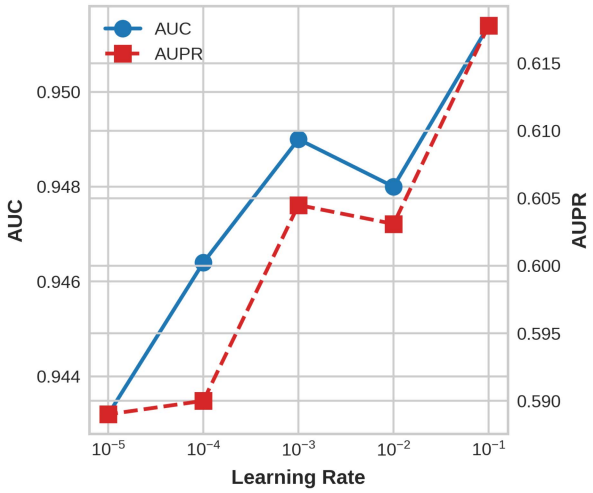


Fig. 2. Impact of different learning rates on model performance.

computing multi-view self-attention. However, overly large values may introduce noise, as confirmed by the poorer performance of FusionMVSA when α was 11 compared to when it was 2. Moreover, when α was set to 7, FusionMVSA outperformed FusionMVSA-A, a variant without similarity-guided terms used in the ablation experiment. The performance of FusionMVSA improved by 0.47% (AUC), 1.41% (AUPR), and 1.17% (F1-score). Therefore, we selected 7 as the optimal value for α. In addition, we tuned the learning rate by testing values from 1×10^{-3} to 5×10^{-3}. As illustrated in Fig. 2, The model achieved its best performance at a learning rate of 1×10^{-3}, which we consequently selected for all our experiments.

C. Model Performance and Compare

To validate the performance of FusionMVSA, we compared it with several existing association prediction methods. The details of these baseline methods are as follows.

- SCPMF takes biological information into account by introducing similarity constraints for drugs and viruses into the probabilistic matrix factorization process [7].
- MKDGRLS calculates multiple kernels from different drug and disease similarities and utilizes related Laplacian regularization terms and a dual Laplacian least squares algorithm to predict associations [45].
- NIMCGCN is a neural inductive matrix completion method with a graph convolutional network used to learn feature representations for predicting miRNA-disease associations [46].
- LAGCN employs a graph convolutional network to learn embeddings on a heterogeneous network, and uses an attention mechanism to combine multi-layer embeddings for drug–disease associations prediction [47].
- MKGCN predicts microbe-drug associations via a method based on multiple kernel fusion on the graph convolutional network [48].
- DRGBCN integrates heterogeneous drug-disease networks through a deep bilinear attention mechanism and layer-wise attention GCN to capture complex interaction patterns for drug repositioning prediction [49].
- LBMFF leverages BERT-based literature semantic features and multi-source information fusion through attention-based GCN to predict novel drug-disease associations [50].
- WMAGT combines Graph Transformer Networks and multi-aggregate GCN to learn heterogeneous graph representations, effectively capturing both local and global node interactions through self-attention mechanisms for drug-disease association prediction [51].
- AutoDR introduces an automatic collaborative learning framework that enhances GNN's message-passing mechanism with neighbor interaction metrics and contrastive learning, enabling more effective feature alignment between disease-drug and neighbor spaces for drug repositioning prediction [52].
- DFDRNN ddresses biased encoding in drug repositioning by mining both similarity and association features from biomedical networks. It utilizes self-attention with single-domain and cross-domain modules to generate more accurate drug and disease representations, demonstrating strong potential for discovering new drug-disease associations [53].

Three types k-fold cross-validation methods, 10-fold, 5-fold, and 2-fold, are applied with Fdataset and Cdataset. As the number of k-folds decreased, the number of known associations

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
PERFORMANCE OF ALL COMPARED METHODS OBTAINED IN 10-FOLD CROSS-VALIDATION ON FDATASET AND CDATASET

Datasets	Methods	AUC	AUPR	Precision	Recall	F1-score
FDataset	LAGCN	0.9077	0.2916	0.3324	0.3864	0.3502
	NIMCGCN	0.8508	0.3806	0.5496	0.3911	0.4543
	SCPMF	0.8961	0.4035	0.5018	0.4128	0.4497
	MKGCN	0.9223	0.5311	0.5793	0.5189	0.5443
	MKDGRLS	0.9365	0.5477	0.5929	0.5111	0.5435
	DRGBCN	0.9310	0.4087	0.4626	0.4325	0.4471
	LBMFF	0.8096	0.1176	0.1734	0.3399	0.2296
	WMAGT	0.9353	0.5231	0.5841	0.5089	0.5473
	AutoDR	0.9473	0.5700	0.4826	**0.5872**	0.5298
	DFDRNN	**0.9543**	0.6100	0.6583	0.5809	0.6104
	FusionMVSA	0.9514	**0.6178**	**0.6661**	0.5799	**0.6192**
CDataset	LAGCN	0.9311	0.3631	0.4357	0.4013	0.4126
	NIMCGCN	0.8623	0.4800	0.6236	0.4621	0.5256
	SCPMF	0.9137	0.4867	0.5900	0.4526	0.5100
	MKGCN	0.9389	0.6365	0.7163	0.5624	0.6277
	MKDGRLS	0.9506	0.6396	0.6952	0.5592	0.6162
	DRGBCN	0.9495	0.4423	0.4548	0.5150	0.4831
	LBMFF	0.9093	0.2026	0.2797	0.4123	0.3333
	WMAGT	0.9458	0.6000	0.7113	0.5279	0.6154
	AutoDR	0.9590	0.6507	0.4873	**0.6829**	0.5688
	DFDRNN	**0.9676**	0.6872	0.7345	0.5990	0.6571
	FusionMVSA	0.9605	**0.6875**	**0.7641**	0.6189	**0.6797**

Note: The best results in each row are in bold faces and the second-best results are underlined.

decreased, thereby gradually increasing the prediction difficulty.The experimental details provided in Table IV demonstrate that FusionMVSA clearly achieves the best AUPR, Recall, and F1-score on both datasets.

The DFDRNN model is a formidable baseline, and its high performance stems from its innovative architecture. It leverages a dual-feature extraction strategy, mining both similarity and association features to encode drugs and diseases. This is accomplished through two specialized modules: the single-domain dual-feature extraction module for intra-domain features and the cross-domain dual-feature extraction module for inter-domain features. Combined with a self-attention mechanism and a cross-dual-domain decoder, these components allow DFDRNN to create robust and comprehensive feature representations.While DFDRNN's approach is powerful, the results indicate that our proposed FusionMVSA model may possess a more refined mechanism for feature fusion or optimization in the context of imbalanced data. The consistent lead in F1-score and AUPR suggests that FusionMVSA more effectively translates its learned representations into precise and reliable predictions. This indicates that FusionMVSA not only generates high-quality encodings but also excels at distinguishing the subtle patterns of true drug-disease associations from the noise inherent in highly imbalanced biomedical networks.

In summary, FusionMVSA outperforms ten recent state-of-the-art association prediction methods on two benchmark datasets, evaluated under 10-fold cross-validation.

D. Ablation Analysis

To identify the contribution of guided terms in computing multi-view self-attention and self-attention, FusionMVSA was compared with three variants using 10-fold cross-validation on the gold-standard Fdataset. The variants are summarized as follows:

- FusionMVSA-A: It lacks the guided terms in the feature extraction component; in other words, topk is set as 0. The aggregate weights of different views are only calculated by the scaled dot-product similarity
- FusionMVSA-B: It ignores the two guided terms, and all aggregate weights are only computed by the scaled dot-product similarity when extracting drug features, disease features, and fusing the two types of features.
- FusionMVSA-C: The guided terms that are in the feature extraction component are retained. The known drug–disease information used as guided terms in the feature fusion component is removed.

TABLE V
PERFORMANCE OF VARIANTS UNDER 10-FOLD CROSS-VALIDATION ON FDATASET

Methods	AUC	AUPR	Precision	Recall	F1-score
FusionMVSA-A	0.9467	0.6037	0.6694	0.5602	0.6075
FusionMVSA-B	0.9420	0.5456	0.6417	0.4950	0.5546
FusionMVSA-C	0.9431	0.5667	0.5908	0.5530	0.5701
FusionMVSA	**0.9514**	**0.6178**	**0.6661**	**0.5799**	**0.6192**

The ROC and PR curves are plotted in Fig. 3 , and the experimental details are shown in Table V . The results demonstrate that all three variants performed worse than FusionMVSA, highlighting the contribution of the guided terms. Furthermore, the guided terms in the feature fusion component play the most critical role in FusionMVSA, as evidenced by FusionMVSA-C performing worse than FusionMVSA-A.

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.

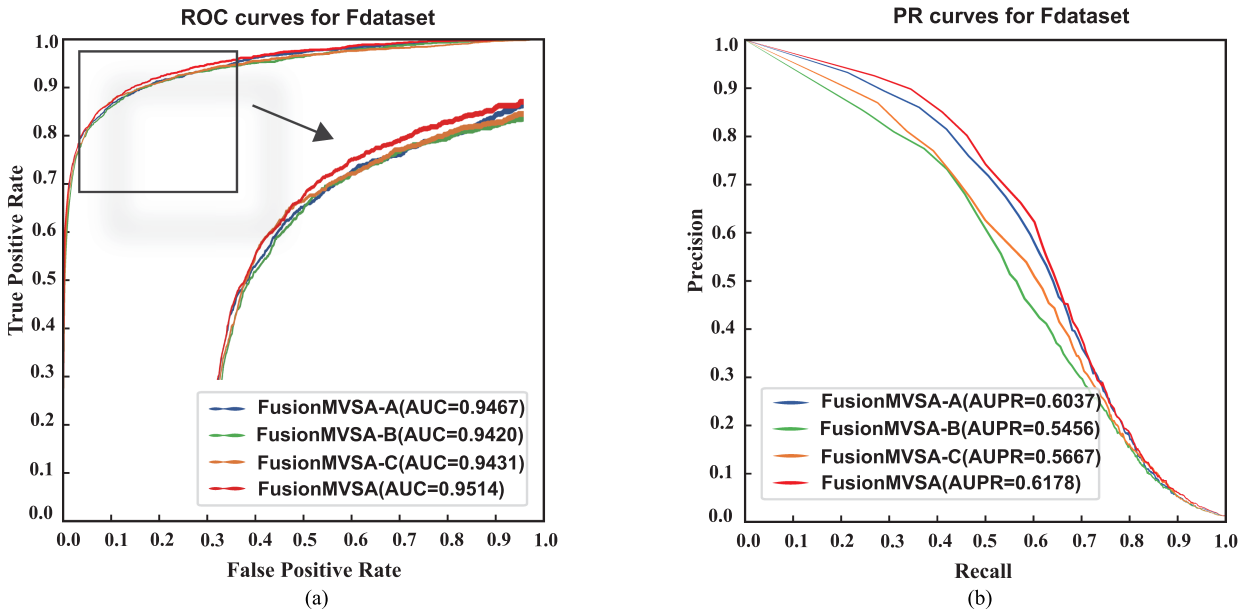


Fig. 3. Performance of FusionMVSA and other three variants in 10-fold cross-validation on Fdataset. (A) Receiver operating characteristic (ROC) curves of FusionMVSA and it's variants. (B) Precision-recall (PR) curves of FusionMVSA and it's variant.

To further investigate the contribution of each component in our model, we not only analyzed the role of the guided terms but also conducted an ablation study to evaluate the contribution of different feature perspectives to the final prediction performance. We designed eight variants, each removing one specific feature view:

- FusionMVSA-woCS&DD: This variant removes the chemical structure similarity (S^R_{CS}) and drug-drug interaction similarity (S^R_{DD}) during fusion.
- FusionMVSA-woCS&DT: This variant removes the chemical structure similarity (S^R_{CS}) and drug-target information similarity (S^R_{DT}) during fusion.
- FusionMVSA-woCS&SE: This variant removes the chemical structure similarity (S^R_{CS}) and side effect similarity (S^R_{SE}) during fusion.
- FusionMVSA-woDD&DT: This variant removes the drug-drug interaction similarity (S^R_{DD}) and drug-target information similarity (S^R_{DT}) during fusion.
- FusionMVSA-woDD&SE: This variant removes the drug-drug interaction similarity (S^R_{DD}) and side effect similarity (S^R_{SE}) during fusion.
- FusionMVSA-woSE&DT: This variant removes the side effect similarity (S^R_{SE}) and drug-target information similarity (S^R_{DT}) during fusion.
- FusionMVSA-woDO: This variant removes the disease ontology similarity (S^D_{DO}) during fusion.
- FusionMVSA-woDP: This variant removes the disease phenotype similarity (S^D_{DP}) during fusion.

The experimental results are presented in Table VI. As shown, the performance of all variants decreased to varying extents compared to the complete FusionMVSA model. This demonstrates that each integrated view provides unique and complementary information for the model. Notably, FusionMVSA-woDP, which removes the disease phenotype similarity, exhibited the most significant drop in performance, with its AUC and AUPR decreasing to 0.9427 and 0.5801, respectively. This indicates that disease phenotypic similarity is crucial for accurately predicting drug-disease associations.

In summary, these results validate the necessity and effectiveness of multi-view feature fusion in our model.

TABLE VI
PERFORMANCE COMPARISON FOR THE ABLATION STUDY ON DIFFERENT FEATURE VIEWS

Methods	AUC	AUPR	Precision	Recall	F1-score
FusionMVSA-woCS&DD	0.9497	0.5927	0.5345	0.5595	0.5640
FusionMVSA-woCS&DT	0.9478	0.5504	0.4162	0.6337	0.5015
FusionMVSA-woCS&SE	0.9492	0.5946	0.5061	0.6192	0.5550
FusionMVSA-woDD&DT	0.9477	0.5537	0.4221	0.6301	0.5047
FusionMVSA-woDD&SE	0.9500	0.5933	0.5148	0.6192	0.5617
FusionMVSA-woSE&DT	0.9498	0.5559	0.4102	0.6513	0.5023
FusionMVSA-woDO	0.9442	0.5908	0.4414	0.5241	0.5302
FusionMVSA-woDP	0.9427	0.5801	0.5093	0.6166	0.5575
FusionMVSA	**0.9514**	**0.6178**	**0.6661**	**0.5799**	**0.6192**

E. Cross-Domain Experiment

To evaluate the generalization ability of our model across different domains, we conducted a cross-domain experiment. Specifically, we trained our model on the Fdataset and tested it on the Cdataset.This experiment aims to test how well the model can adapt to a new, unseen dataset after training on the source domain, thereby assessing its robustness and ability to transfer learned knowledge across domains. We compared our model's performance with that of two state-of-the-art models. The results are summarized in the Table VII. The comparative analysis reveals that our model achieved promising results,

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
CROSS-DOMAIN EXPERIMENT RESULTS

Methods	AUC	AUPR	Precision	Recall	F1-score
AutoDR	0.8521	0.4633	0.3805	0.5126	0.4368
DFDRNN	0.8795	0.5109	0.5217	0.4688	0.4938
FusionMVSA	0.8714	0.5183	0.5492	0.4815	0.5131

TABLE VIII
IMPACT OF DIFFERENT NEGATIVE SAMPLE RATIO STRATEGIES ON THE TEN-FOLD CROSS-VALIDATION RESULTS

Metric	Ratio 1:1	Ratio 1:2	Ratio 1:3	Ratio 1:4	Ratio 1:5
AUC	0.9251	0.9340	0.9381	0.9418	0.9429
AUPR	0.9265	0.8911	0.8681	0.8545	0.8344
Precision	0.8710	0.8482	0.8209	0.8004	0.8002
Recall	0.8427	0.7884	0.7848	0.7584	0.7305
F1-score	0.8564	0.8166	0.8020	0.7781	0.7635

demonstrating substantial generalization ability and robustness when applied to an unseen dataset.

F. Impact of Different Negative Sample Ratio Strategies

We further investigated the impact of different negative sample selection strategies on the experimental results. The current assumption that all unobserved drug-disease pairs are considered as negative samples may introduce noise into the model. Specifically, unobserved pairs may represent potential positive interactions that are simply not recorded, leading to inaccurate negative sample labeling. We experimented with various negative sample ratios, ranging from 1:1 to 1:5, and evaluated the model performance using ten-fold cross-validation. The results, summarized in Table VIII , demonstrate that increasing the ratio of negative samples results in a decline in key metrics, including AUPR, Precision, Recall, and F1-score. This suggests that higher negative sample ratios may introduce more noise and reduce the model's ability to correctly identify true positive interactions.

Despite these challenges, our model consistently performed well across all ratios, achieving high AUC and reasonable Precision-Recall trade-offs. These findings highlight the limitations of the current negative sample selection strategy and underscore the need for more refined methods to select negative samples, such as hard-negative mining, to avoid the introduction of noise and further enhance model performance.

G. Case Studies

To further verify the reliability of FusionMVSA in drug recommendation, we conducted case studies on computationally predicted drug candidates for 10 neoplasms. All known drug–disease associations were used as the training set, while all unknown pairs were treated as the candidate set.

Neoplasms are serious conditions often treated with a combination of surgery and drugs. The drug–disease associations predicted by our method were verified using highly reliable sources and clinical trials, including the Comparative Toxicogenomics Database (CTD), DrugBank, and ClinicalTrials. The CTD is a powerful open-source research resource providing vital chemical–disease association information. DrugBank and ClinicalTrials are authoritative sources for clinical trial information. We first selected therapeutic drug–disease associations (noted as CTD_T in the CTD as evidence. If no evidence was found that the predicted drug could treat the disease, we searched for clinical trial studies in DrugBank or ClinicalTrials. Tables X and XI lists the prediction results and corresponding evidence for the Fdataset and Cdataset, respectively. Among the 50 predictions, 28 have therapeutic evidence in the CTD, and only 3 of the remaining 22 predictions have no evidence. Complete lists of all predicted drug–disease associations for the two datasets are provided with the FusionMVSA code.

Additionally, we conducted molecular docking experiments to further validate our model. Using Multiple Myeloma as an example, we calculated the molecular binding energies between the top five candidate drugs predicted by FusionMVSA on the Fdataset and five target proteins associated with the disease. These interactions were analyzed using Autodock Vina and Discovery Studio (DS) software to elucidate the protein-ligand binding modes. For the unvalidated drug Dacarbazine's effect on Multiple Myeloma, we selected Cereblon (PDB code: 8aoq) as the target protein. The molecular binding energy between Dacarbazine and Cereblon was found to be -5.6 kcal mol (Table IX). Fig. 4 illustrates the van der Waals interactions between the

TABLE IX
DOCKING ENERGIES OF THE TOP 5 CANDIDATE DRUGS PREDICTED BY FUSIONMVSA FOR MULTIPLE MYELOMA WITH FIVE TARGET PROTEINS

Drug	2fne	3grw	4zfo	6fs1	8aoq
Cytarabine	-6.5	-5.8	-6.0	-5.7	-5.7
Methotrexate	-5.9	-7.6	-7.1	-6.7	-7.3
Fludarabine	-5.9	-6.1	-7.0	-5.4	-6.1
Dacarbazine	-5.1	-5.5	-5.3	-5.0	-5.6
Mitoxantrone	-7.0	-6.1	-7.0	-6.3	-6.4

TABLE X
TOP 5 FUSIONMVSA-PREDICTED DRUGS FOR 5 NEOPLASMS ON FDATASET

Disease (OMIM ID)	Drug	DrugBank ID	Evidence
Colorectal Neoplasms (114500)	Doxorubicin	DB00997	CTD_T
	Amifostine	DB01143	DrugBank
	Cisplatin	DB00515	CTD_T
	Lapatinib	DB01259	DrugBank
	Carboplatin	DB00958	DrugBank
Esophageal Neoplasms (133239)	Doxorubicin	DB00997	CTD_T
	Mitomycin	DB00305	CTD_T
	Leucovorin	DB00650	DrugBank
	Capecitabine	DB01101	CTD_T
	Temozolomide	DB00853	CTD_T
Stomach Neoplasms (137215)	Cisplatin	DB00515	CTD_T
	Carboplatin	DB00958	DrugBank
	Irinotecan	DB00762	CTD_T
	Busulfan	DB01008	N/A
	Lapatinib	DB01259	CTD_T
Lung Neoplasms (211980)	Cisplatin	DB00515	DrugBank
	Methotrexate	DB00563	CTD_T
	Carboplatin	DB00958	CTD_T
	Mitomycin	DB00305	CTD_T
	Dacarbazine	DB00851	DrugBank
Multiple Myeloma (254500)	Cytarabine	DB00987	CTD_T
	Methotrexate	DB00563	CTD_T
	Fludarabine	DB01073	DrugBank
	Dacarbazine	DB00851	N/A
	Mitoxantrone	DB01204	ClinicalTrials

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.

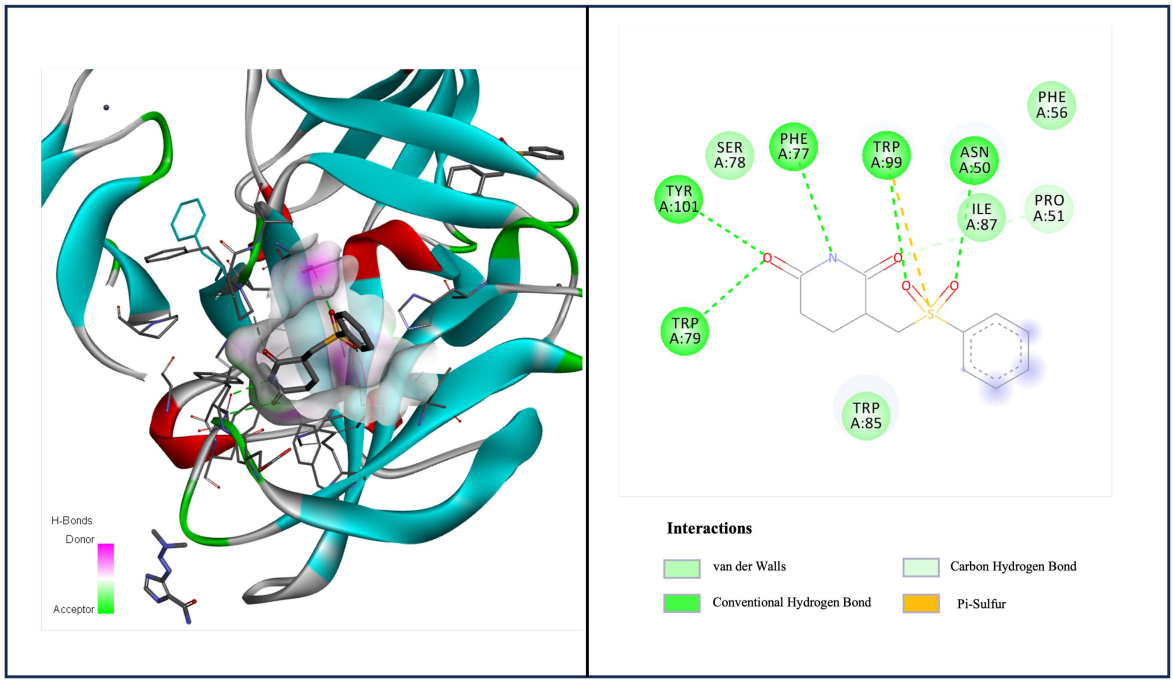


Fig. 4. Docked Dacarbazine (DrugBank ID: DB00851) with (PDB code:8AOQ) as well as their interactions.

TABLE XI
TOP 5 FUSIONMVSA-PREDICTED DRUGS FOR 5 NEOPLASMS ON CDATASET

Disease (OMIM ID)	Drug	DrugBank ID	Evidence
Urinary Bladder Neoplasms (109800)	Vincristine	DB00541	ClinicalTrials
	Leucovorin	DB00650	CTD_T
	Capecitabine	DB01101	ClinicalTrials
	Docetaxel	DB01248	CTD_T
	Irinotecan	DB00762	DrugBank
Carcinoma, Hepatocellular (114550)	Sunitinib	DB01268	DrugBank
	Gemcitabine	DB00441	CTD_T
	Hydroxyurea	DB01005	ClinicalTrials
	Regorafenib	DB08896	CTD_T
	Fluorouracil	DB00544	CTD_T
Glioma (137800)	Cisplatin	DB00515	CTD_T
	Hydroxyurea	DB01005	ClinicalTrials
	Melphalan	DB01042	CTD_T
	Methotrexate	DB00563	CTD_T
	Pamidronate	DB00282	N/A
Carcinoma, Renal Cell (144700)	Docetaxel	DB01248	DrugBank
	Paclitaxel	DB01229	DrugBank
	Lapatinib	DB01259	DrugBank
	Doxorubicin	DB00997	CTD_T
	Fluorouracil	DB00544	CTD_T
Prostatic Neoplasms (176807)	Methotrexate	DB00563	CTD_T
	Paclitaxel	DB01229	CTD_T
	Doxorubicin	DB00997	CTD_T
	Vinblastine	DB00570	CTD_T
	Lapatinib	DB01259	DrugBank

small molecule and specific amino acid residues (SER78, ILE87, TRP85, PHE56). Additionally, various intermolecular interactions were observed, including conventional hydrogen bonds between nitrogen, oxygen, sulfur atoms and residues (TRP79, TYR101, PHE77, TRP99, ASV50), as well as carbon-hydrogen bond interactions between an oxygen atom and PRO51. These experiments further validate the effectiveness of the FusionMVSA method in predicting potential therapeutic drugs for new diseases. They also reveal the specific interaction mechanisms between drugs and target proteins, providing critical insights for new drug development.

V. DISCUSSION AND CONCLUSION

In this study, we propose a multi-view information fusion, self-attention-based deep learning methodology, termed FusionMVSA, for predicting drug–disease associations.

Unlike previous methods that rely on heterogeneous information integration for association prediction, FusionMVSA leverages both the distinct and shared characteristics of each multi-view dataset through a novel multi-view fusion strategy with self-attention. This mechanism enhances traditional self-attention by allowing feature learning to follow a dynamic model guided by specific terms. Our approach begins with a similarity-based feature extraction component designed to capture multi-view representations of drugs and diseases. These representations are then automatically fused through a feature fusion component, where known drug–disease associations act as guiding terms. This process enables the exploration of complementary properties across different views. Finally, rather than employing linear multiplication of latent features, we utilize multi-layer neural networks to capture complex associations. Comparative evaluations on two datasets using 10-fold cross-validation demonstrate that FusionMVSA significantly outperforms state-of-the-art association prediction approaches and various FusionMVSA variants. This robust performance underscores FusionMVSA's efficacy in identifying repurposable drug candidates across multiple drug–disease networks.

Despite the effectiveness of FusionMVSA in predicting drug -disease associations, there are some limitations to this approach. Firstly, the reliability of known drug–disease associations is questionable, as many of these associations are derived from prediction tools that may contain numerous inaccuracies. To

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.

enhance prediction performance, FusionMVSA could benefit from incorporating more comprehensive and updated data, utilizing substantially related biological information. Secondly, the current knowledge of drugs and diseases is insufficient. Future work will aim to integrate both unstructured databases and structured ontologies across various domains—including genomics, molecular biology, healthcare, and therapeutics—to construct a comprehensive knowledge graph.

To summarize, our proposed FusionMVSA methodology focuses on modeling the probability of drug–disease associations by incorporating a similarity-guided multi-view self-attention mechanism with a multi-layer perceptron neural network. This approach leverages the advantages of multi-view representations, considering both the commonalities and distinctiveness of each element within the multi-view data. Ultimately, FusionMVSA will not only aid in identifying new candidate compounds with potential therapeutic effects for experimental validation but also contribute to the prevention, diagnosis, and treatment of diseases.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data supporting this research can be accessed at the following link: https://github.com/Biowust/FusionMVSA

References

[1] J. W. Scannell, A. Blanckley, H. Boldon, and B. Warrington, "Diagnosing the decline in pharmaceutical r&d efficiency," *Nature Rev. Drug Discov.*, vol. 11, no. 3, pp. 191–200, 2012.

[2] S. Pushpakom et al., "Drug repurposing: Progress, challenges and recommendations," *Nature Rev. Drug Discov.*, vol. 18, no. 1, pp. 41–58, 2019.

[3] A. Hamosh, A. F. Scott, J. S. Amberger, C. A. Bocchini, and V. A. McKusick, "Online mendelian inheritance in man (omim), a knowledgebase of human genes and genetic disorders," *Nucleic Acids Res.*, vol. 33, no. suppl_1, pp. D514–D517, 2005.

[4] D. S. Wishart et al., "Drugbank: A comprehensive resource for in silico drug discovery and exploration," *Nucleic Acids Res.*, vol. 34, no. suppl_1, pp. D668–D672, 2006.

[5] S. Kim et al., "Pubchem substance and compound databases," *Nucleic Acids Res.*, vol. 44, no. D1, pp. D1202–D1213, 2016.

[6] S. Avram et al., "Drugcentral 2021 supports drug discovery and repositioning," *Nucleic Acids Res.*, vol. 49, no. D1, pp. D1160–D1169, 2021.

[7] Y. Meng, M. Jin, X. Tang, and J. Xu, "Drug repositioning based on similarity constrained probabilistic matrix factorization: Covid-19 as a case study," *Appl. Soft Comput.*, vol. 103, 2021, Art. no. 107135.

[8] V. Martínez, C. Navarro, C. Cano, W. Fajardo, and A. Blanco, "Drugnet: Network-based drug–disease prioritization by integrating heterogeneous data," *Artif. Intell. Med.*, vol. 63, no. 1, pp. 41–49, 2015.

[9] H. Luo et al., "Drug repositioning based on comprehensive similarity measures and bi-random walk algorithm," *Bioinformatics*, vol. 32, no. 17, pp. 2664–2671, 2016.

[10] L. Peng, M. Ren, L. Huang, and M. Chen, "GEnDDn: An lncRNA–disease association identification framework based on dual-net neural architecture and deep neural network," *Interdiscipl. Sci.: Comput. Life Sci.*, vol. 16, no. 2, pp. 418–438, 2024.

[11] L. Peng, L. Huang, Q. Su, G. Tian, M. Chen, and G. Han, "LDA-VGHB: Identifying potential lncRNA–disease associations with singular value decomposition, variational graph auto-encoder and heterogeneous newton boosting machine," *Brief. Bioinf.*, vol. 25, no. 1, 2024, Art. no. bbad466.

[12] J. Wei et al., "Enhancing drug–food interaction prediction with precision representations through multilevel self-supervised learning," *Comput. Biol. Med.*, vol. 171, 2024, Art. no. 108104.

[13] J. Wei, L. Zhuo, S. Pan, X. Lian, X. Yao, and X. Fu, "Headtailtransfer: An efficient sampling method to improve the performance of graph neural network method in predicting sparse ncrna–protein interactions," *Comput. Biol. Med.*, vol. 157, 2023, Art. no. 106783.

[14] Z. Zhou et al., "GAM-MDR: Probing miRNA–drug resistance using a graph autoencoder based on random path masking," *Brief. Funct. Genomic.*, vol. 23, no. 4, pp. 475–483, 2024.

[15] Z. Zhou et al., "MHAM-NPI: Predicting ncRNA-protein interactions based on multi-head attention mechanism," *Comput. Biol. Med.*, vol. 163, 2023, Art. no. 107143.

[16] Z. Zhou, L. Zhuo, X. Fu, J. Lv, Q. Zou, and R. Qi, "Joint masking and self-supervised strategies for inferring small molecule-miRNA associations," *Mol. Ther.-Nucleic Acids*, vol. 35, no. 1, 2024.

[17] Z. Zhou, L. Zhuo, X. Fu, and Q. Zou, "Joint deep autoencoder and subgraph augmentation for inferring microbial responses to drugs," *Brief. Bioinf.*, vol. 25, no. 1, 2024, Art. no. bbad483.

[18] R. Wang et al., "An effective plant small secretory peptide recognition model based on feature correction strategy," *J. Chem. Inf. Model.*, vol. 64, no. 7, pp. 2798–2806, 2023.

[19] L. Zhuo, R. Wang, X. Fu, and X. Yao, "Stablednam: Towards a stable and efficient model for predicting DNA methylation based on adaptive feature correction learning," *BMC Genomic.*, vol. 24, no. 1, 2023, Art. no. 742.

[20] Y. Wang, Y. Zhai, Y. Ding, and Q. Zou, "Sbsm-pro: Support bio-sequence machine for proteins," *Sci. China Inf. Sci.*, vol. 67, no. 11, 2024, Art. no. 212106.

[21] A. Gottlieb, G. Y. Stein, E. Ruppin, and R. Sharan, "Predict: A method for inferring novel drug indications with application to personalized medicine," *Mol. Syst. Biol.*, vol. 7, no. 1, 2011, Art. no. 496.

[22] X. Liang et al., "LRSSL: Predict and interpret drug–disease associations based on data integration using sparse subspace learning," *Bioinformatics*, vol. 33, no. 8, pp. 1187–1196, 2017.

[23] W. Zhang et al., "Predicting drug-disease associations by using similarity constrained matrix factorization," *BMC Bioinf.*, vol. 19, pp. 1–12, 2018.

[24] M. Yang, L. Huang, Y. Xu, C. Lu, and J. Wang, "Heterogeneous graph inference with matrix completion for computational drug repositioning," *Bioinformatics*, vol. 36, no. 22-23, pp. 5456–5464, 2020.

[25] Z. Wang, M. Zhou, and C. Arnold, "Toward heterogeneous information fusion: Bipartite graph convolutional networks for in silico drug repurposing," *Bioinformatics*, vol. 36, no. Supplement_1, pp. i525–i533, 2020.

[26] L. Cai et al., "Drug repositioning based on the heterogeneous information fusion graph convolutional network," *Brief. Bioinf.*, vol. 22, no. 6, 2021, Art. no. bbab319.

[27] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Representations*, 2017. [Online]. Available: https://openreview.net/forum?id=SJU4ayYgl

[28] Y. Mourchid and R. Slama, "D-STGCNT: A dense spatio-temporal graph conv-GRU network based on transformer for assessment of patient physical rehabilitation," *Comput. Biol. Med.*, vol. 165, 2023, Art. no. 107420.

[29] P. Jaccard, "Nouvelles recherches sur la distribution florale," *Bull. Soc. Vaud. Sci. Nat.*, vol. 44, pp. 223–270, 1908.

[30] M. Babaei, T. M. Evers, F. Shokri, L. Altucci, E. C. de Lange, and A. Mashaghi, "Biochemical reaction network topology defines dose-dependent drug–drug interactions," *Comput. Biol. Med.*, vol. 155, 2023, Art. no. 106584.

[31] D. Weininger, "Smiles, a chemical language and information system. 1. introduction to methodology and encoding rules," *J. Chem. Inf. Comput. Sci.*, vol. 28, no. 1, pp. 31–36, 1988.

[32] C. Steinbeck, Y. Han, S. Kuhn, O. Horlacher, E. Luttmann, and E. Willighagen, "The chemistry development kit (cdk): An open-source java library for chemo-and bioinformatics," *J. Chem. Inf. Comput. Sci.*, vol. 43, no. 2, pp. 493–500, 2003.

[33] M. Kuhn, I. Letunic, L. J. Jensen, and P. Bork, "The sider database of drugs and side effects," *Nucleic Acids Res.*, vol. 44, no. D1, pp. D1075–D1079, 2016.

[34] M. A. Van Driel, J. Bruggeman, G. Vriend, H. G. Brunner, and J. A. Leunissen, "A text-mining analysis of the human phenome," *Eur. J. Hum. Genet.*, vol. 14, no. 5, pp. 535–542, 2006.

[35] C. E. Lipscomb, "Medical subject headings (mesh)," *Bull. Med. Library Assoc.*, vol. 88, no. 3, pp. 265–266, 2000.

[36] Q. Li, Z. Han, and X.-M. Wu, "Deeper insights into graph convolutional networks for semi-supervised learning," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1.

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.

[37] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. 27th Int. Conf. Mach. Learn.*, 2010, pp. 807–814.
[38] A. Vaswani, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017.
[39] J. L. Ba, "Layer normalization," 2016, *arXiv:1607.06450*.
[40] S. Narang et al., "Do transformer modifications transfer across implementations and applications?," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, M.-F. Moens, X. Huang, L. Specia, and S. W.-T. Yih, Eds., Nov. 2021, pp. 5758–5773, doi: 10.18653/v1/2021.emnlpmain.465. [Online]. Available: https://aclanthology.org/2021.emnlp-main.465/
[41] K. Zou, Z. Wang, S. Zhu, S. Wang, and F. Yang, "IDRNet: A novel pixel-enlightened neural network for predicting protein subcellular location based on interactive pointwise attention," *Curr. Bioinf.*, vol. 18, no. 10, pp. 805–816, 2023.
[42] C. Ying et al., "Do transformers really perform badly for graph representation?," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 28877–28888.
[43] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.
[44] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Representations*, 2015, San Diego, CA, USA, May 7-9, 2015, Conf. Track Proc., Y. Bengio and Y. LeCun, Eds., 2015. [Online]. Available: http://arxiv.org/abs/1412.6980
[45] H. Yang, Y. Ding, J. Tang, and F. Guo, "Drug–disease associations prediction via multiple kernel-based dual graph regularized least squares," *Appl. Soft Comput.*, vol. 112, 2021, Art. no. 107811. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1568494621007328
[46] C. Jin, Z. Shi, K. Lin, and H. Zhang, "Predicting miRNA-disease association based on neural inductive matrix completion with graph autoencoders and self-attention mechanism," *Biomolecules*, vol. 12, no. 1, 2022. [Online]. Available: https://www.mdpi.com/2218-273X/12/1/64
[47] B.-W. Zhao, L. Hu, Z.-H. You, L. Wang, and X.-R. Su, "Hingrl: Predicting drug–disease associations with graph representation learning on heterogeneous information networks," *Brief. Bioinf.*, vol. 23, no. 1, Dec. 2021, Art. no. bbab515, doi: 10.1093/bib/bbab515.
[48] H. Yang, Y. Ding, J. Tang, and F. Guo, "Inferring human microbe–drug associations via multiple kernel fusion on graph neural network," *Knowl.-Based Syst.*, vol. 238, 2022, Art. no. 107888. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0950705121010546
[49] X. Tang et al., "Enhancing drug repositioning through local interactive learning with bilinear attention networks," *IEEE J. Biomed. Health Informat.*, vol. 29, no. 3, pp. 1644–1655, Mar. 2025.
[50] H. Kang, L. Hou, Y. Gu, X. Lu, J. Li, and Q. Li, "Drug–disease association prediction with literature based multi-feature fusion," *Front. Pharmacol.*, vol. 14, 2023, Art. no. 1205144.
[51] S. He, L. Yun, and H. Yi, "Fusing graph transformer with multi-aggregate GCN for enhanced drug–disease associations prediction," *BMC Bioinf.*, vol. 25, no. 1, 2024, Art. no. 79.
[52] Y. Wang et al., "Automatic collaborative learning for drug repositioning," *Eng. Appl. Artif. Intell.*, vol. 139, 2025, Art. no. 109653.
[53] E. Zhu, X. Li, C. Liu, and N. R. Pal, "Boosting drug-disease association prediction for drug repositioning via dual-feature extraction and cross-dual-domain decoding," *J. Chem. Inf. Model.*, vol. 65, no. 11, pp. 5745–5757, 2025.

Authorized licensed use limited to: Beijing Jiaotong University. Downloaded on October 01,2026 at 01:06:12 UTC from IEEE Xplore. Restrictions apply.